BTW Sandpile

Steady-State Mean Height $\langle z \rangle_\infty$

From simulation to spanning trees

Bak, Tang & Wiesenfeld (1987) · **Priezzhev (1994)**

The puzzle: simulation vs. naive estimate

Simulation

$$\langle z \rangle_{sim} \approx 2.100$$

100×100 grid, steady state

Heights $z \in \{0, 1, 2, 3\}$

Open boundary conditions

One grain added per step

≠

+40%

Naive estimate

$$\langle z \rangle_{naive} = 1.5$$

Assumption: uniform distribution

over all values {0, 1, 2, 3}

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(0 + 1 + 2 + 3) / 4 = 1.5
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✗ *Not all configs. are reachable*

Bak, Tang & Wiesenfeld, Phys. Rev. Lett. 59, 381 (1987)

Recurrent configurations — Majumdar & Dhar (1992)

Steady state

The stationary distribution is uniform over recurrent configurations $\mathcal{R}$ — not over all 4^{N^2} possible configs.

Forbidden configs. (FSC)

A subset S of sites is forbidden if every site $i \in S$ has $z(i) <$ number of neighbours of i in S. No config. in $\mathcal{R}$ contains an FSC.

Effect on heights

Patterns with many z=0 or z=1 tend to be FSC → excluded from $\mathcal{R}$. Distribution shifts toward z=2,3.

The no-FSC condition systematically excludes low-height configurations → ⟨z⟩ rises well above 1.5

FSC — Formal definition

Definition (Forbidden Sub-Configuration)

A subset $S \subseteq \Lambda$ is an FSC if and only if

$$\forall\, i \in S : \quad z(i) < |\{ j \in S : j \sim i \}|$$

where $j \sim i$ denotes lattice adjacency.

In words: every site in S has fewer grains than the number of its neighbours that lie inside S.

A configuration is recurrent ⟺ it contains no FSC.

Burning test (Dhar 1990)

1. Add a sink node at the boundary.
2. Burn sites iteratively: burn i if z(i) ≥ unburnt neighbours.
3. All burnt → recurrent ✓
4. Remaining S unburnable → S is FSC ✗

Dhar, Phys. Rev. Lett. 64, 1613 (1990) · Majumdar & Dhar, Physica A 185, 129 (1992)

FSC — Examples on the 2D lattice (z_crit = 4)

Example 1

S = { i }, z(i) = 0

2	2	2
2	0	2
2	2	2

|{j ∈ S : j~i}| = 0
z(i) = 0 < 0 ? NO

✓ Not an FSC — isolated site at 0 has no neighbours in S

Example 2

S = { i, j }, z(i)=z(j)=0

2	2	2
2	0	0
2	2	2

|{j∈S:j~i}| = 1 for both
z(i)=0 < 1 ✓
z(j)=0 < 1 ✓

✗ Minimal FSC on the 2D lattice

Example 3

S = 2×2 cluster, z=1

1	1	3
1	1	3
3	3	3

Corner sites: 1 neighbour in S
Edge sites: 2 neighbours in S
z=1 < 2 ✓ for edge sites

✗ Connected cluster is forbidden

On the 2D lattice each bulk site has 4 neighbours; boundary sites have 2 or 3

Spanning trees & Green's function — Priezzhev (1994)

Fundamental bijection

$\mathcal{R} \longleftrightarrow$ **spanning trees of the graph**

with sink node at boundary (Dhar's theorem)

The count $|\mathcal{R}|$ is computed via the Kirchhoff matrix-tree theorem:

$$|\mathcal{R}| = \det \Delta$$

Δ = discrete Laplacian (Dirichlet boundary)

Average via Green's function

The mean $\langle z \rangle$ over $\mathcal{R}$ is obtained by differentiating log det Δ w.r.t. an auxiliary field.

The result involves Catalan's constant:

$$G \approx 0.9159656 \quad (\Sigma\ (-1)^k/(2k+1)^2)$$

Bulk height distributions:

P(z=1) P(z=2) P(z=3)

all expressible in terms of G/π^2

Priezzhev, J. Stat. Phys. 74, 955 (1994) · Dhar, Phys. Rev. Lett. 64, 1613 (1990)

Exact result and comparison with simulation

$$\langle z \rangle_\infty \approx 2.125$$

thermodynamic limit $N \to \infty$ (Priezzhev, 1994)

Naive estimate

1.500

uniform distribution
over {0,1,2,3}

✗ *ignores $\mathcal{R}$*

Priezzhev (1994)

2.125

spanning trees +
Green's function

✓ *exact ($N \to \infty$)*

Simulation

2.100

100×100 grid
steady state

~ *finite-size effects*

The gap 2.125 → 2.100 reflects $O(1/N)$ corrections: boundary sites have $\langle z \rangle$ systematically lower than the bulk value.

Levels of depth — from statistical physics to combinatorics

#	Approach	Result	Difficulty	Note
1	Conservation argument	$\langle z \rangle > 0$, stationary	*trivial*	*does not fix $\langle z \rangle$*
2	Uniform dist. over {0..3}	$\langle z \rangle = 1.5$	*trivial*	*wrong: ignores $\mathcal{R}$*
3	No-FSC condition (Majumdar-Dhar)	$\langle z \rangle > 1.5$, pushed upward	*medium*	*qualitative*
4	Spanning trees + Green's function	$\langle z \rangle_\infty \approx 2.125$	*advanced*	*exact (Priezzhev 1994)*
5	Numerical simulation	$\langle z \rangle \approx 2.100$	—	*finite size N=100*

Dhar (1990) · Majumdar & Dhar (1992) · Priezzhev (1994)